Customer Churn Prediction Using Machine Learning

William Chen

Abstract

This project investigates customer churn prediction in the telecom industry using Logistic Regression and Random Forest. The dataset contains over 500,000 customer records with features including support call frequency, payment delay, contract length, and total spend. Model performance is evaluated using Accuracy, Sensitivity, Specificity, F1-score, and AUC-ROC.

Initial results reveal a significant discrepancy between training accuracy (89.5%) and test accuracy (57.1%), with a specificity of only 19.4%. Through distribution analysis, we identify a distributional inconsistency between the original train and test sets. Following a random re-split, Logistic Regression achieves a test accuracy of 84.8% with an AUC of 0.908, while Random Forest achieves 93.6% accuracy and an AUC of 0.953. Feature selection using BIC removed `Tenure` with no impact on performance, and coefficient analysis identified `Contract.Length` (Monthly odds ratio = 8.10) and `Support.Calls` (odds ratio = 1.50) as the strongest churn predictors.

1 Introduction

Customer churn refers to the rate at which customers discontinue their subscription or cease using a company's products and services within a given period. For telecom companies, churn is a critical business metric, as retaining existing customers is substantially more cost-effective than acquiring new ones. Identifying customers at risk of churning enables companies to intervene proactively, through targeted retention strategies, personalized offers, or service improvements, before the customer decides to leave.

This project investigates customer churn prediction on a telecom customer dataset of over 500,000 records using Logistic Regression and Random Forest. Beyond predictive modeling, this work emphasizes identifying and addressing data quality issues that affect model reliability. Through exploratory data analysis, we uncover a distribution shift in the original train/test split and demonstrate that correcting the data partitioning substantially improves generalization performance. We further compare feature selection methods (AIC and BIC) and evaluate whether non-linear models offer improved predictive accuracy over logistic regression.

2 Data

2.1 Dataset Overview

The dataset contains 505,207 customer records sourced from Kaggle, representing telecom subscribers over a 12-month observation period. The dataset is pre-divided into a training set (440,833 observations) and a test set (64,374 observations). During preprocessing, one incomplete observation was removed from the training set, resulting in a final training set of 440,832 records. The target variable *Churn* is binary, where 1 indicates a customer who discontinued their subscription and 0 indicates a retained customer. The training set contains 190,833 non-churned (43.3%) and 249,999 churned (56.7%) customers, indicating a mild class imbalance.

2.2 Feature Description

The dataset contains 11 variables: 1 identifier (CustomerID, excluded from analysis), 1 binary target (Churn), and 10 predictors. Features span demographic (Age, Gender), be-

havioral (Usage Frequency, Support Calls, Payment Delay, Total Spend, Last Interaction), and contractual (Tenure, Subscription Type, Contract Length) dimensions. Among the 10 predictors, 3 are categorical and 7 are continuous.

3 Exploratory Data Analysis

3.1 Summary Statistics

All continuous features fall within expected ranges, with no extreme outliers identified. One incomplete observation was detected in the training set and removed prior to analysis, resulting in a final training set of 440,832 records.

3.2 Feature Distribution by Churn

Figure 1 shows the overall churn distribution in the training set. Of 440,832 observations, 190,833 (43.3%) are non-churned and 249,999 (56.7%) are churned, indicating a mild class imbalance that is unlikely to substantially affect model performance.

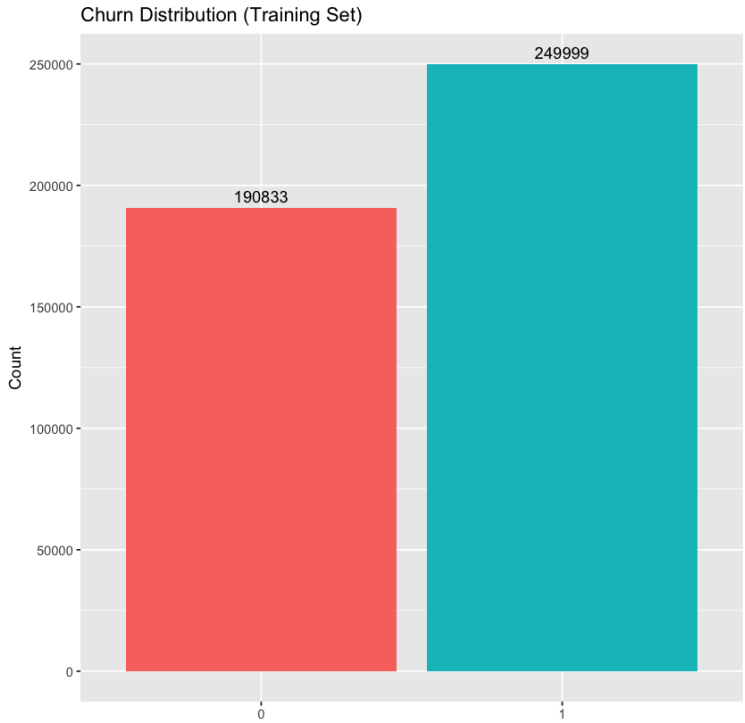

Figure 1: Churn Distribution (Training Set)

Figure 2 shows the churn rate across categorical features. Contract Length shows the most notable pattern: customers on Monthly contracts exhibit a substantially higher churn rate compared to those on Quarterly or Annual plans. This suggests that the absence of a long-term commitment is strongly associated with churn risk. Gender and Subscription Type show relatively balanced churn distributions across categories, indicating these features may have limited discriminative power on their own.

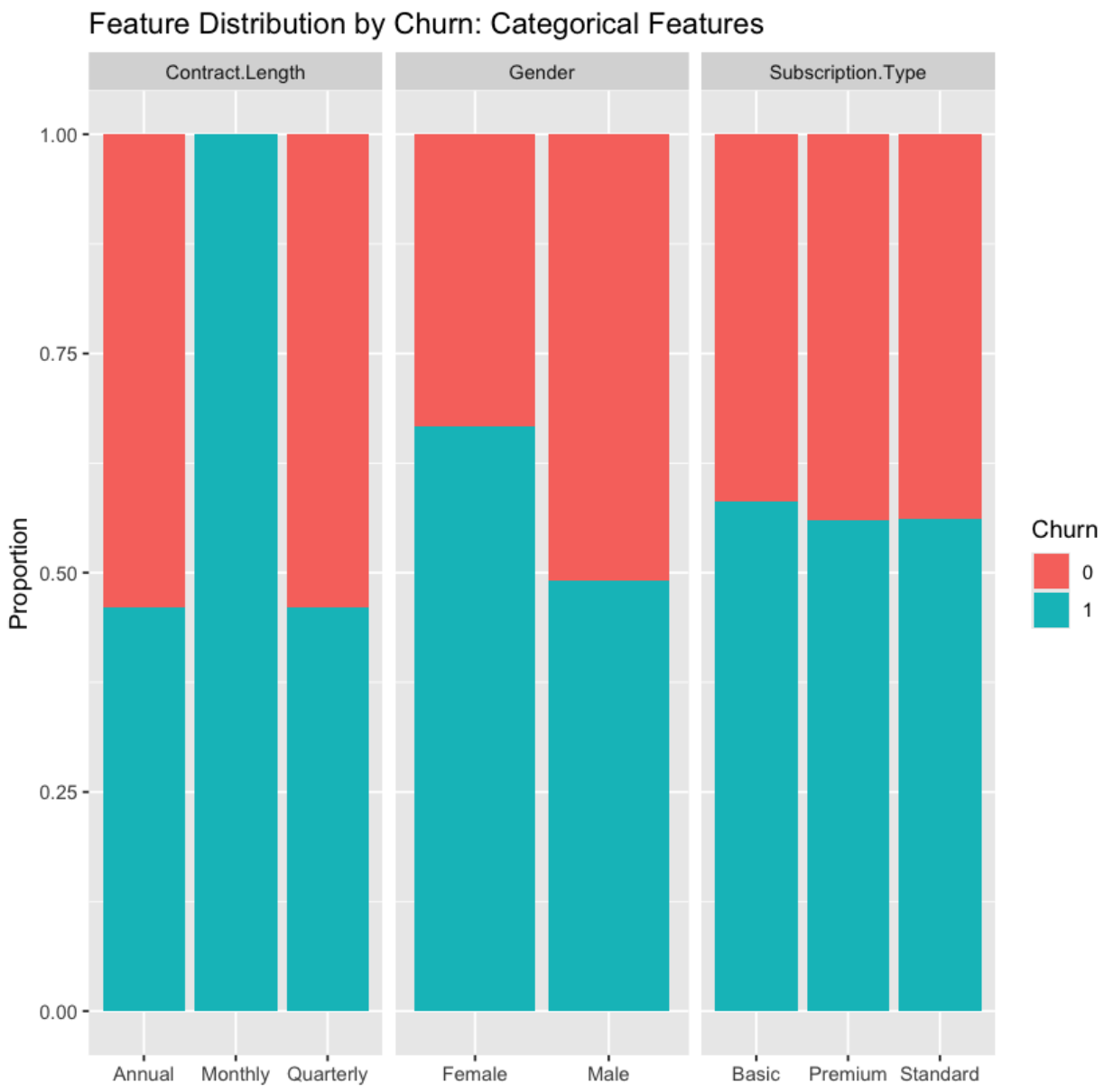

Figure 2: Feature Distribution by Churn: Categorical Features

Figure 3 shows the distribution of numeric features by churn status. Support Calls shows the most pronounced difference between churned and non-churned customers, with churned customers making substantially more calls. This suggests that higher support call frequency reflects customer dissatisfaction and is a strong indicator of churn risk. Other features such as Tenure and Usage Frequency show minimal differences between the two groups.

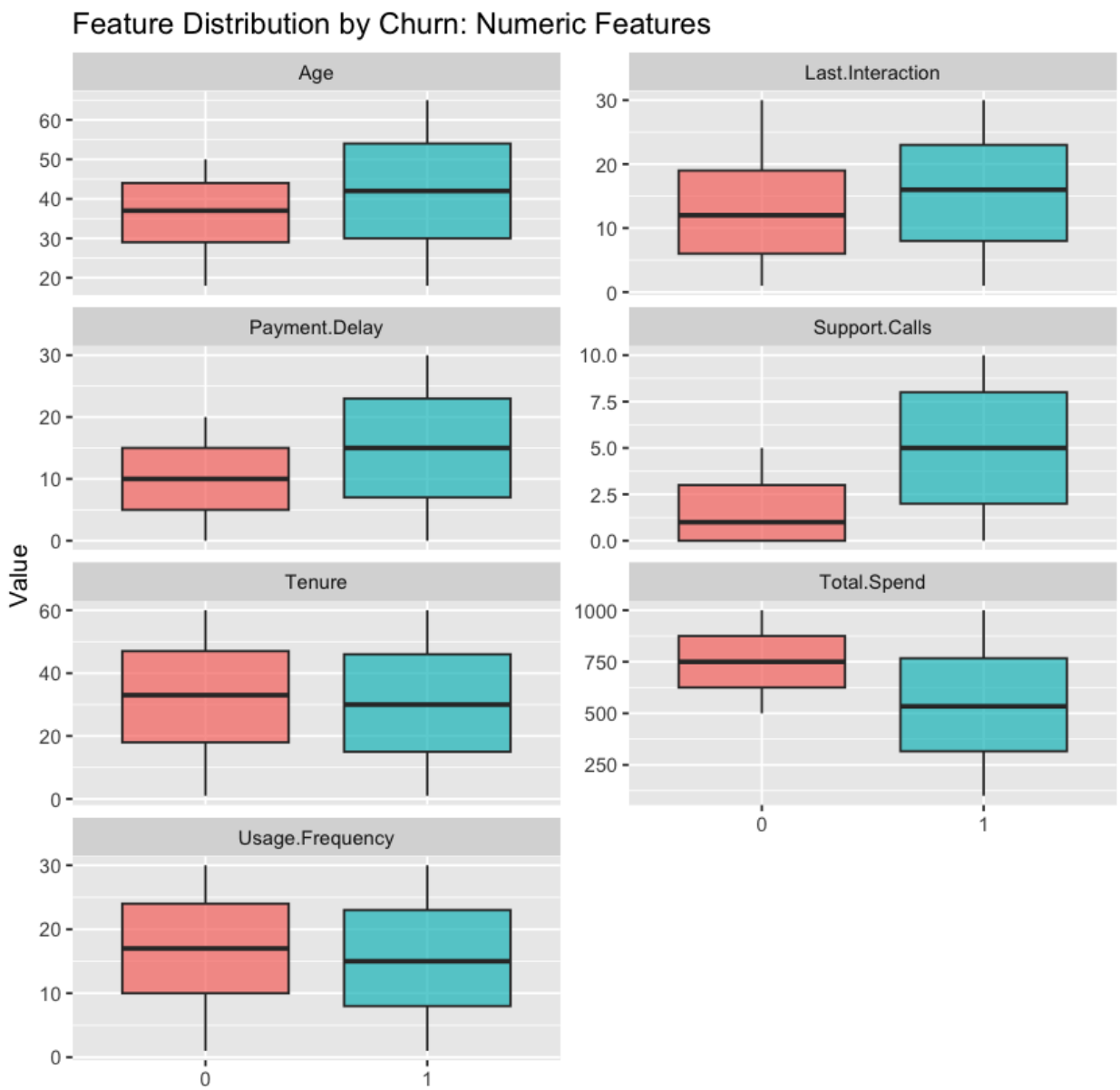

Figure 3: Feature Distribution by Churn: Numeric Features

3.3 Feature Correlation

Support Calls shows the strongest positive correlation with Churn ($r = 0.57$), consistent with the pattern observed in Section 3.2, confirming that higher support call frequency is a strong indicator of churn risk. Total Spend shows the strongest negative correlation ($r = -0.43$), suggesting that customers with higher spending are less likely to leave. This is intuitive: loyal, high-value customers tend to be more engaged with the service, while customers who

are already considering leaving may have reduced their usage and spending before eventually churning. The remaining features show weak to moderate correlations with Churn.

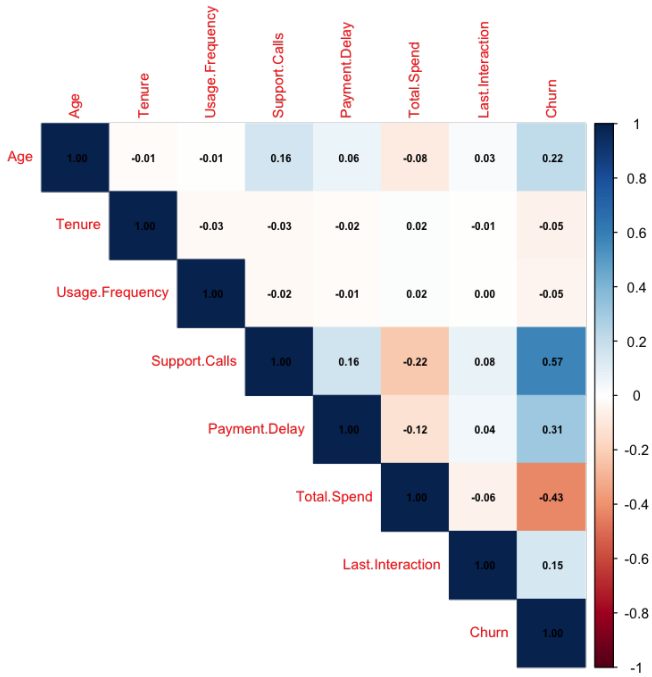

Figure 4: Feature Correlation Matrix

3.4 Multicollinearity Check (VIF)

To assess multicollinearity among predictors, Variance Inflation Factors (VIF) were computed for all features. VIF measures the extent to which each feature can be predicted by the remaining features, with values above 10 generally indicating problematic collinearity. All features return VIF values below 1.15, well within the acceptable range, confirming that the predictors are largely independent of one another and that all features can be retained for modeling.

4 Methodology

4.1 Feature Selection

Backward elimination was applied to identify relevant predictors. Two selection criteria were compared: AIC ($k = 2$) and BIC ($k = \log(n)$). AIC penalizes model complexity with a fixed cost per parameter, while BIC applies a stricter penalty that scales with sample size, making it more conservative in large datasets. With over 400,000 observations, BIC provides a more rigorous test of whether each variable contributes meaningfully to prediction.

On the original training set, AIC retained all 10 predictors. After re-splitting the data, AIC again retained all predictors, while BIC removed Tenure, reducing the model to 9 predictors. This result is consistent with the exploratory analysis, where Tenure showed minimal distributional difference between churned and non-churned customers.

4.2 Machine Learning Models

Logistic Regression Logistic Regression was selected as the primary model for its interpretability, providing coefficient estimates and odds ratios that translate directly into actionable business insights. Two variants were trained: a full model retaining all 10 predictors (AIC), and a reduced model with 9 predictors after BIC selection.

Random Forest Random Forest was included as a non-linear benchmark to evaluate whether the linear assumptions of logistic regression limit predictive performance. Due to computational constraints, the model was trained on a random subsample of 50,000 observations from the re-split training set. All trees use the full set of 10 predictors, as Random Forest internally handles variable importance through its splitting mechanism.

4.3 Evaluation Metrics

Model performance was assessed using Accuracy, Sensitivity, Specificity, F1-score, and AUC-ROC. These metrics were chosen to capture different aspects of classification performance, particularly given the asymmetric costs of misclassifying churners versus non-churners in a business context.

5 Results

5.1 Logistic Regression (Original Split)

The logistic regression model achieved a cross-validated training accuracy of 89.5%, but test accuracy dropped to 57.1%. Sensitivity was 99.1% while Specificity was only 19.4%, indicating the model predicted nearly all observations as churned, misclassifying the majority of non-churners.

To investigate, density plots were generated for the two most influential predictors—Support Calls and Total Spend—stratified by churn label across train and test sets.

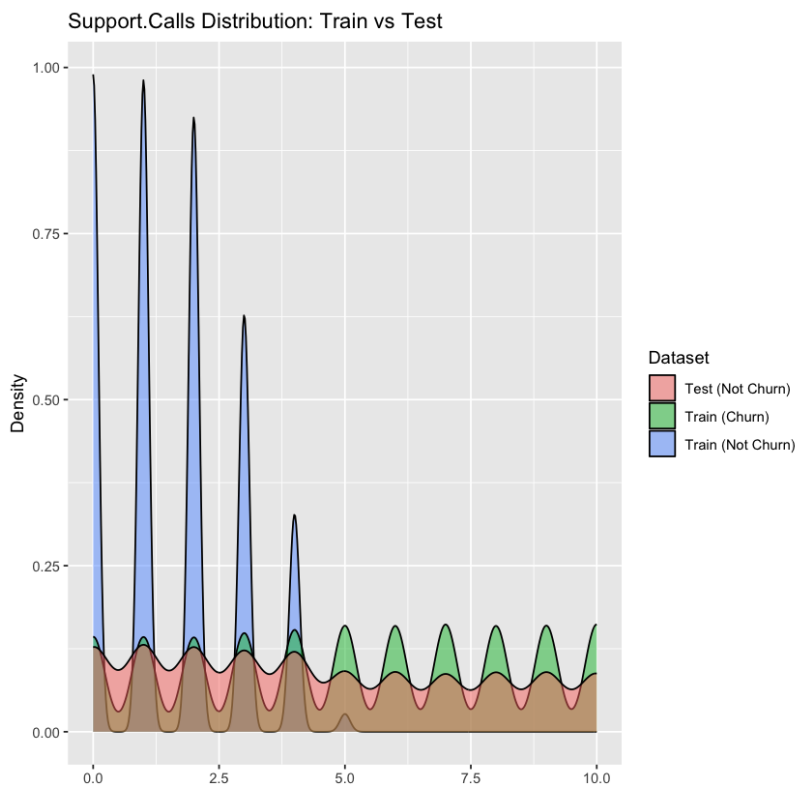

Figure 5: Density plot of Support Calls by churn label across train and test sets.

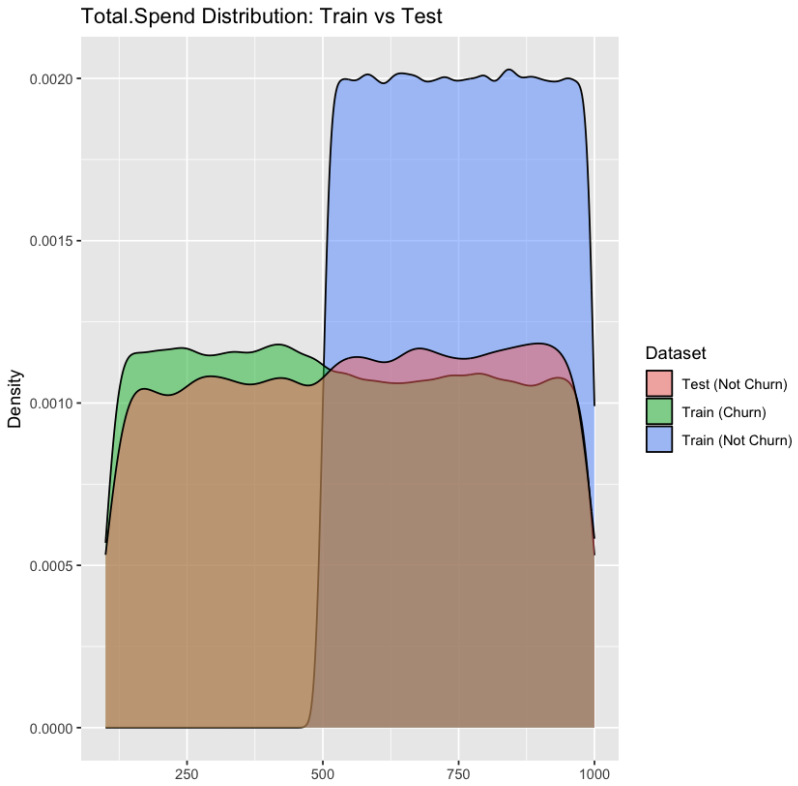

Figure 6: Density plot of Total Spend by churn label across train and test sets.

Both figures reveal that the non-churned group in the test set follows a markedly different distribution from its training-set counterpart, overlapping substantially with the training set's churned group. This confirms a distributional inconsistency in the original data partition, and that the poor test performance reflects a data quality issue rather than a model deficiency.

5.2 Re-split: Logistic Regression

After merging and randomly re-splitting the data (80/20), two logistic regression models were trained. The AIC model retained all 10 predictors, while the BIC model removed Tenure, reducing to 9 predictors.

Table 1: Logistic Regression: AIC vs. BIC (Re-split)

	Accuracy	**Sensitivity**	**Specificity**	**F1**	**AUC**
AIC (10 features)	0.8475	0.8478	0.8470	0.8609	0.9084
BIC (9 features)	0.8475	0.8478	0.8472	0.8609	0.9083

Removing Tenure had negligible impact on performance. The close alignment between cross-validated accuracy (84.9%) and test accuracy (84.8%) indicates stable generalization with no overfitting.

Coefficient Analysis Table 2 presents the full coefficient estimates and odds ratios from the AIC logistic regression model.

Table 2: Logistic Regression Coefficients and Odds Ratios (AIC Model)

Feature	**Coefficient**	**Odds Ratio**	**95% CI**	**P-Value**
(Intercept)	−1.287	0.276	[0.262, 0.291]	< 0.001
Age	0.023	1.023	[1.023, 1.024]	< 0.001
Gender: Male	**−0.727**	**0.483**	[0.475, 0.492]	< 0.001
Tenure	−0.001	0.999	[0.999, 1.000]	0.001
Usage Frequency	−0.013	0.987	[0.986, 0.988]	< 0.001
Support Calls	**0.406**	**1.500**	[1.495, 1.506]	< 0.001
Payment Delay	0.084	1.088	[1.086, 1.089]	< 0.001
Subscription: Premium	−0.078	0.925	[0.905, 0.945]	< 0.001
Subscription: Standard	−0.069	0.933	[0.913, 0.953]	< 0.001
Contract Length: Monthly	**2.092**	**8.102**	[7.872, 8.338]	< 0.001
Contract Length: Quarterly	−0.010	0.990	[0.972, 1.009]	0.314
Total Spend	−0.003	0.997	[0.997, 0.997]	< 0.001
Last Interaction	0.033	1.033	[1.032, 1.034]	< 0.001

5.3 Re-split: Random Forest

Random Forest was trained on a 50,000-observation subsample.

Table 3: Random Forest Results (Re-split)

Accuracy	**Sensitivity**	**Specificity**	**F1**	**AUC**
0.9357	0.9968	0.8590	0.9452	0.9529

Random Forest outperformed logistic regression across all metrics, with Accuracy improving by approximately 9 percentage points and AUC by 4.5 percentage points, suggesting that the linear assumptions of logistic regression constrain its predictive capacity on this dataset.

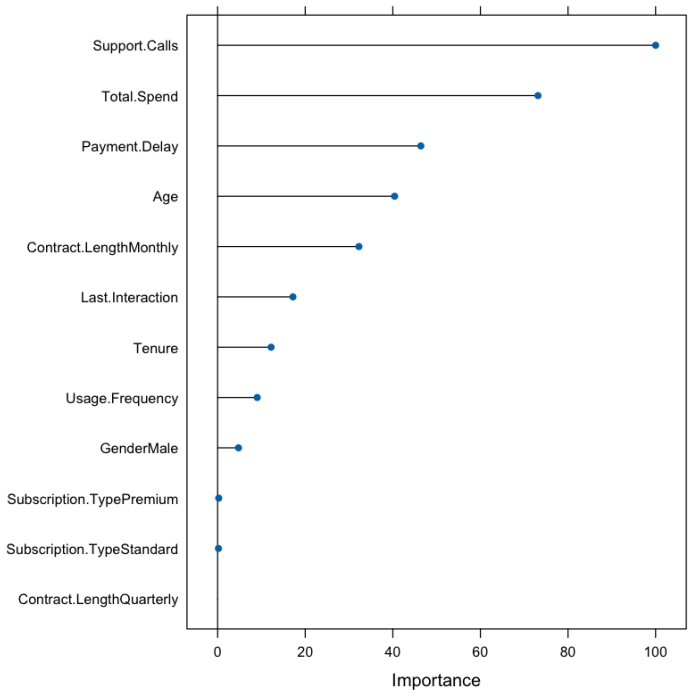

Figure 7: Variable Importance from Random Forest Model

Variable Importance Support Calls, Total Spend, and Payment Delay are the top three predictors, consistent with the logistic regression coefficient analysis. Tenure ranks in the lower half, aligning with the BIC result that excluded it.

5.4 Overall Comparison

Table 4: Overall Model Comparison

Model	Accuracy	Sensitivity	Specificity	F1	AUC
LR Original	0.5712	0.9906	0.1938	0.6864	0.6831
LR AIC (Re-split)	0.8475	0.8478	0.8470	0.8609	0.9084
LR BIC (Re-split)	0.8475	0.8478	0.8472	0.8609	0.9083
RF (Re-split)	0.9357	0.9968	0.8590	0.9452	0.9529

6 Discussion & Conclusion

6.1 Key Findings

The logistic regression coefficient analysis identified `Contract.Length`, `Support.Calls`, and `Gender` as the features with the largest effect sizes. Monthly contract holders exhibited the highest churn risk with an odds ratio of 8.10 relative to Annual customers, each additional `Support.Calls` increased churn odds by 50%, and male customers showed approximately half the churn odds of female customers. The Random Forest variable importance ranking highlighted `Support.Calls`, `Total.Spend`, and `Payment.Delay` as the top predictors. While the two models rank features differently due to their distinct modeling assumptions, `Support.Calls` consistently emerged as a top predictor across both approaches.

Beyond predictive performance, this analysis uncovered a distributional inconsistency between the original train and test sets. This highlights the importance of verifying data quality before modeling: applying a pre-defined split without examining distributional consistency can lead to misleading evaluations, regardless of how well the model itself is specified.

6.2 Business Recommendations

`Contract.Length` exhibited the largest effect on churn, with Monthly subscribers showing 8.10 times the churn odds of Annual customers. Offering incentives such as discounts or added benefits to encourage upgrades to Quarterly or Annual plans could substantially reduce churn risk. Notably, Quarterly and Annual customers showed no significant difference in churn behavior ($p = 0.314$), suggesting that any longer-term commitment is effective.

`Support.Calls` was consistently identified as a top predictor by both logistic regression (odds ratio = 1.50) and Random Forest (highest variable importance). Each additional support call increased churn odds by 50%, suggesting that high call volumes reflect unresolved customer issues. Improving first-contact resolution and proactively addressing recurring complaints may help retain at-risk customers.

Female customers exhibited approximately twice the churn odds of male customers, suggesting a need for targeted retention efforts for this segment.

6.3 Limitations

While Random Forest outperformed logistic regression across all metrics, confirming that non-linear relationships exist in the data, it comes at the cost of interpretability. The logistic regression model remains more suitable for deriving actionable business insights through its coefficient estimates and odds ratios. Additionally, the Random Forest was trained on a subsample of 50,000 observations due to computational constraints, which may not fully represent the complete dataset.

Second, the dataset is cross-sectional, providing only a snapshot of customer behavior at a single point in time. Without longitudinal data, it is not possible to track how individual customer behavior evolves over time or to identify early warning signs of churn before they fully manifest.